Reconstructing Japan's Scenic Past from Prints: Combining Citizen Science and AI-Methods for Authenticating Direct Observation in Ukiyo-e Landscapes

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This short paper presents an experimental project bridging past and present through the integration of citizen science and artificial intelligence in analyzing early-modern Japanese visual heritage. Conducted between August 2024 and January 2025 by former Nippon Foundation funding recipients across North America, Europe and Asia, the collaboration incorporated expertise in digital anthropology, East Asian art history, knowledge modeling and comparative Japanese and global histories. The project is scheduled for consolidation and platform implementation throughout 2025 as part of a Postdoctoral Project at the University of Zurich's Chair of East Asian Art History.

The research addresses a fundamental question in Japanese art history: how did *ukiyo-e* illustrators balance direct observation, secondary sources, and cultural-artistic conventions? Rather than assuming they adhered strictly to either approach, our project aims to identify patterns illuminating this dynamic relationship through recognition algorithms, GIS mapping, and citizen science-enhanced spatial analysis.

Japanese "landscape" appears across various Tokugawa-period woodblock-printed works, prompting critical investigations into the term's ontological and technical nature (see: Chino 2003 / Karatani 1993 / Takeuchi 1989). These works include the popular print series Thirty-six Views of Mt. Fuji (1830–1832) by Katsushika Hokusai (1760–1849) or The Fifty-three Stations of the Tōkaidō (first edition 1832) by Utagawa Hiroshige (1797–1858), as well as illustrated books, and sketches about topographical and literary spaces (Smith 2016 / Liagkou et al. 2023). Scholarly discourse continues seeking ways to understand how observational practices, literary sources, and artistic conventions manifest in *ukiyo-e* . While established art-historical methods have provided valuable insights, they can be enhanced by computational approaches engaging with technical and cultural dimensions.

The project thus applies three innovative technical components: Automated Feature Detection using fine-tuned vision language models for Japanese landscape element classification—trained on ca. 3,000 annotated *ukiyo-e* images from the MET to recognize topographical features; Systematic Viewpoint Analysis implementing GIS-based georeferencing workflows with a citizen science component (adapted from Smapshot)—allowing participants to match historical prints to modern topography through persistent landscape features; and Comparative Spatial Analysis contextualizing selected prints within a broader corpus of visual and textual sources to distinguish observation from artistic convention. Central to this project's alignment with open science principles is its integration of citizen science through a public-facing platform combining contribution workspaces with educational resources [Figure 1]. This approach recognizes that cultural significance emerges not only from technical analysis but through cultural interpretation respecting Japanese aesthetic traditions. The platform ensures accessibility through multilingual interfaces (Japanese and English), clear documentation, and interactive tutorials guiding users through the contribution process.

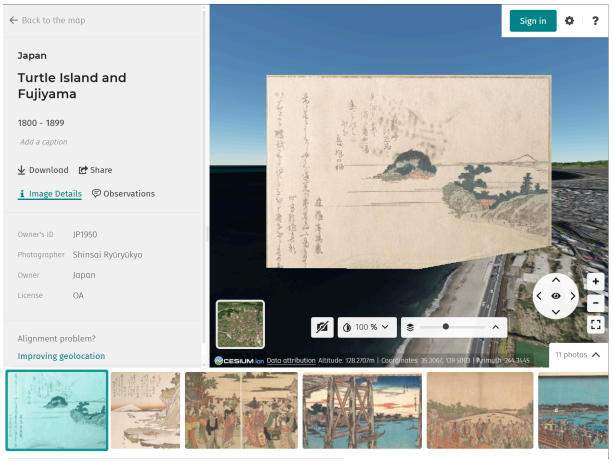

Figure 1 Interactive georeferencing of Enoshima's Turtle Island in the Smapshot project prototype

By triangulating between spatial knowledge of the crowd, historical documents, and digital elevation models built from contemporary topographical data, we establish ground truths that are both technical and art-historical. Initial testing was performed on approximately 3,000 open-access images from the Metropolitan Museum of Art. For example, analysis of Hokusai's "Turtle Island and Mount Fuji" (c. 1830s) revealed an intricate blending of mythological elements with accurate geographical features, demonstrating how *ukiyo-e* illustrators navigated between empirical observation and cultural-literary traditions. Going forward, the project plans to include selected examples from over 200,000 prints from the Digital Archive Research Center at Ritsumeikan University, Kyoto.

Early results suggest that computational methods can identify patterns in landscape representation across large datasets while facilitating public participation in cultural heritage research. By understanding these features and their temporal evolution, we can gain insight into artistic shifts which, when contextualized in the commercial print market of the late Tokugawa period, illuminate societal cultural viewing processes. The research provides a reproducible framework for using vision language models at the intersection of art history and citizen science while remaining sensitive to the cultural and aesthetic dimensions that shaped these works within their specific historical contexts. Landscape transformation documentation, cultural heritage preservation, educational resource compilation, and tourism are likely beneficiaries of this research, conducted open-access wherever possible. Through this approach, the project demonstrates how international collaboration, advancements in vision language models, and citizen participation can enhance both academic and public understanding of historical visual culture.

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